Q: You are working on an NLP model. So, you are dealing with words and sentences, not numbers. Your problem is to categorize these words and make sense of them. Your manager told you that you have to use embeddings.
Which of the following techniques are not related to embeddings? Explain the other terms related to embedding.
A. Count Vector
B. TF-IDF Vector
C. Co-Variance Matrix

Ans: Covariance matrix is not an embedding technique. Covariance matrices are square matrices with the covariance between each pair of elements. It measures how much the change of one with respect to another is related.

Other two are embedding techniques:
Count Vector

- It is one of the simplest ways of doing text vectorization.
- It creates a document term matrix, which is a set of dummy variables that indicates if a particular word appears in the document.
- Count vectorizer will fit and learn the word vocabulary and try to create a document term matrix in which the individual cells denote the frequency of that word in a particular document, which is also known as term frequency, and the columns are dedicated to each word in the corpus.

Example:
Document-1: He is a smart boy. She is also smart.
Document-2: Chirag is a smart person.

The dictionary created contains the list of unique tokens(words) present in the corpus
Unique Words: ['He', 'She', 'smart', 'boy', 'Chirag', 'person'] Here, D=2, N=6
So, the count matrix M of size 2 X 6 will be represented as –

	He	She	smart	boy	Chirag	person
D1	1	1	2	1	0	0
D2	0	0	1	0	1	1

TF-IDF
Term frequency denotes the frequency of a word in a document. For a specified word, it is defined as the ratio of the number of times a word appears in a document to the total number of words in the document.

$$TF(\boldsymbol{term}) = \frac{Number\ of\ times\ \boldsymbol{term}\ appears\ in\ a\ document}{Total\ number\ of\ items\ in\ the\ document}$$

Inverse document frequency looks at how common (or uncommon) a word is amongst the corpus. It measures the importance of the word in the corpus.

Document Frequency tells us about the proportion of documents that contain a certain word. IDF is the reciprocal of the Document Frequency.

$$IDF = log\left(\frac{Total\ number\ of\ documents}{Documents\ containing\ word\ W}\right)$$

The intuition behind using IDF is that the more common a word is across all documents, the lesser its importance is for the current document. A logarithm is taken to dampen the effect of (normalize) IDF in the final calculation.

The final TF-IDF score comes out to be:

$$TF_IDF = TF * IDF$$